CEBT II REACTION ENGINEERING

3. Heterogeneous catalysts

What is catalysis?

First described by Humphry Davy 1817 – "*the discovery of a new and curious series of phenomena*" – the oxidation of coal gas over metal wires, whilst working on his safety lamp.

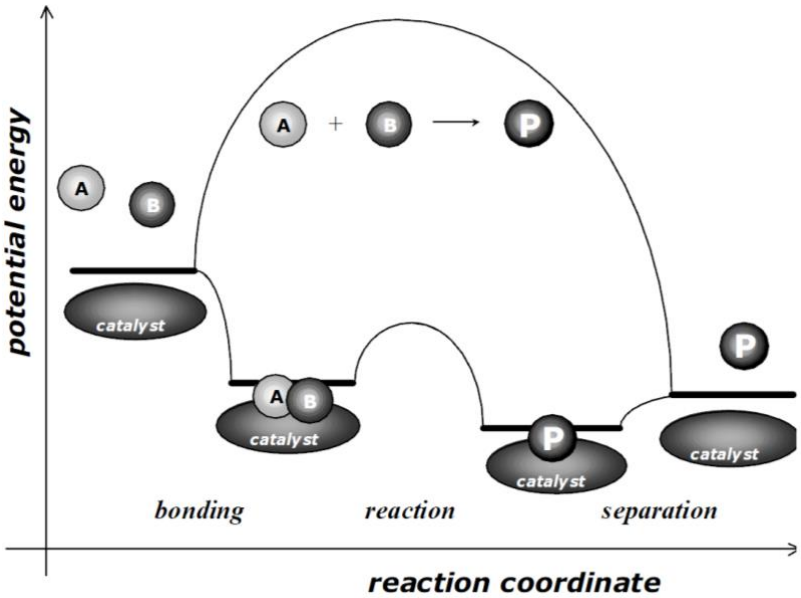
The term "catalysis" was introduced by Berzelius in 1835 as the ability of substances "*to awaken affinities, which are asleep at a particular temperature, by their mere presence and not by their own affinity*".

Modern definition: *Catalysts are materials which change the rate of attainment of chemical equilibrium without themselves being changed or consumed in the process.*

A result of this can also be to provide selectivity to particular products which are more desirable than others.

The equilibrium position is not changed.

The potential energy description:



There is often an endothermic step in product desorption, meaning that transport of the product away from the active site can be as important as the transport of the reactant(s) to it.

Why is catalysis important?

Data from 2020 [M.K. Lynch "UK Catalysis Market Study Summary" for KTN, Innovate UK]:
Value of catalytic materials produced: $34 billion, of which:

Refining 24%

Petrochemicals 20%

Fine chemicals 6%

Environmental 37% (car catalysts)

Polymers 13%

Value of materials produced using these: $3.3 trillion.

By some estimates catalytic processes are essential for ~30% of GDP [Catlow et al., Phil Trans A 374 (2016) 20150089], including >85% of products of the chemical industry.

Nobel Prizes for heterogeneous catalysis:

1903: Svante August Arrhenius "*for advancement of chemistry by his electrolytic theory of dissociation*"

1909: Wilhelm Ostwald "*in recognition of his work on catalysis*"

1912: Paul Sabatier "*for his method of hydrogenating organic compounds in the presence of finely disintegrated metals*"

1918: Fritz Haber "*for the synthesis of ammonia from its elements*"

1931: Carl Bosch "*for the invention and development of chemical high pressure methods*"

1932: Irving Langmuir "*for his discoveries and investigations in surface chemistry*"

1963: Ziegler and Natta "*technology of high polymers*"

2007: Gerhard Ertl "*for his studies of chemical processes on solid surfaces*"

"*Chemistry without catalysis, would be a sword without handle, a light without brilliance, a bell without sound."* Alwin Mittasch (worked for Bosch at BASF, discovered cheap Fe catalyst which makes process possible – originally Os catalyst was used)

Types of catalysis:

Biocatalysis: The catalyst is an enzyme

Homogeneous catalysis: Catalyst is in the same phase as the reactant

Heterogeneous catalysis: (1) catalysis with the catalyst in a different phase than the reactants, and (2) usually with the catalyst on solid, porous material or impregnated in such a material.

There is plenty of room for improvement in industrial catalysts e.g. compare nitrogen fixation by enzymes in plants to the Haber process.

Advantages of heterogeneous catalysis:

Easy separation of product and catalysts

High catalyst lifetimes

Particularly acid catalysis: fewer corrosion and disposal problems

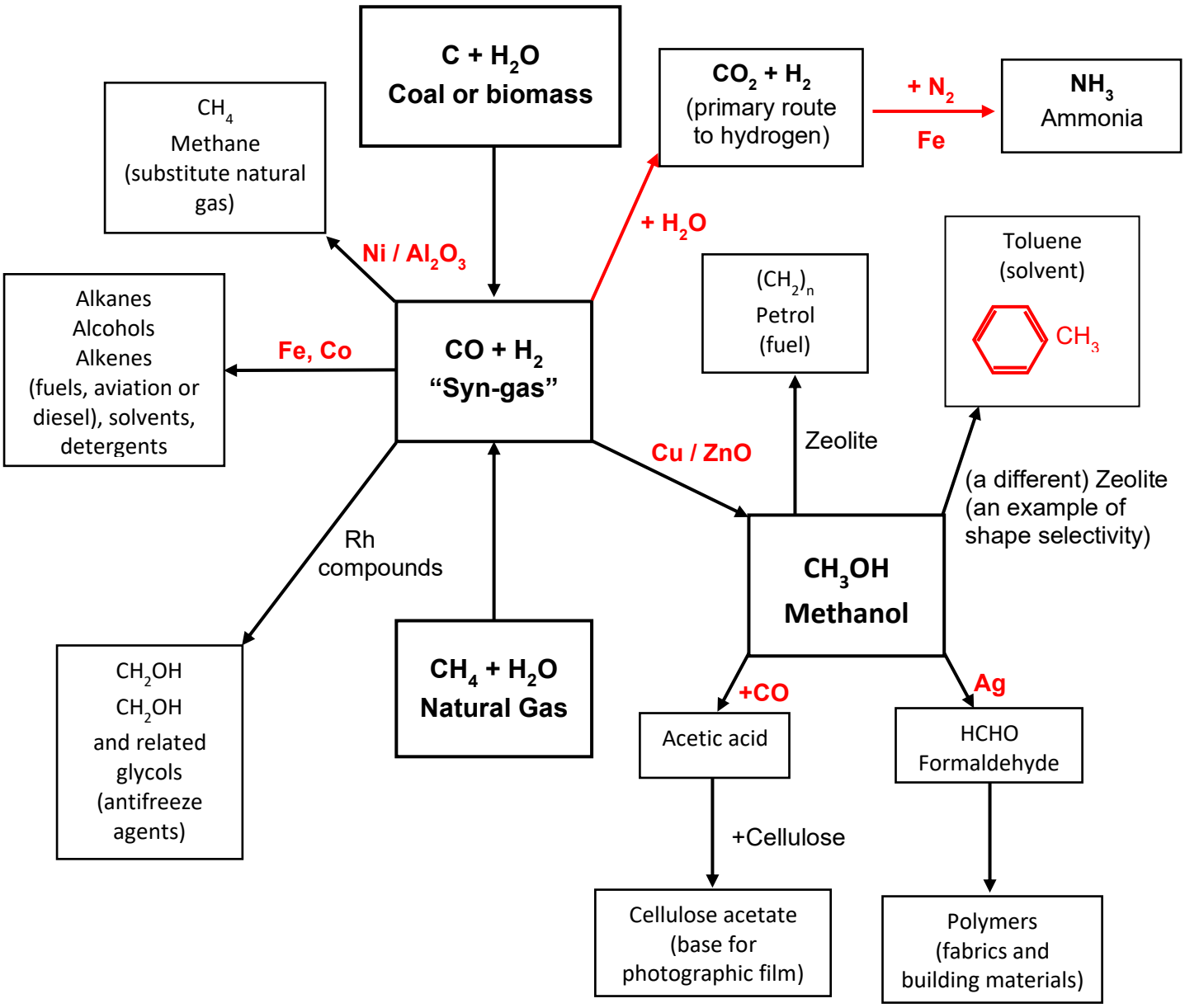
→ ~ 80 % of industrial catalytic processes are heterogeneous

Major Heterogeneous Catalytic Processes:

Process	**Catalyst**
Catalytic cracking of crude oil	Zeolites
Hydrotreating of crude oil	Co-Mo, Ni-Mo, Ni-W
Reforming of naptha to gasoline	Pt, Pt-Re, Pt-Ir
Alkylation	H_2O_4, HF, solid acids
Polymerisation of ethylene and propylene	Cr, $TiCl_x/MgCl_2$
Ethylene epoxidation	Ag
Vinyl chloride synthesis	Cu (as chloride)
Steam reforming of methane to syn-gas	Ni
Water-gas shift	Fe (oxide), Cu-ZnO
Methanation	Ni
Ammonia synthesis	Fe
Ammonia oxidation	Pt-Rh
Acrylonitrile synthesis	Bi-Mo, Fe-Sb (oxides)
Hydrogenation of vegetable oils	Ni
Sulpuric acid manufacture	V (oxide)
Oxidation of CO and hydrocarbons (car exhaust)	Pt, Pd
Reduction of NO_x (car exhaust)	Rh, VO_x

N.b. Significance of transition metals, which are often expensive. C.f enzymes which also often use transition metals.

e.g. the uses of Syn-gas: a platform chemical



Note: changing catalyst and reaction conditions results in a very wide range of products from the same reactants. This illustrates the enormous power of catalysts.

3.1 Types of solid catalyst

- There are many types of solid catalysts covering a wide variety of reactions.
- In all cases, the active ingredient of the catalyst may be present with a binder (e.g. a porous support, or a clay) to give larger particles.
  - Fine free-flowing powders (e.g. zeolite-based cracking catalysts).
  - Spherical beads or cylindrical pellets, typically with dimensions of 2-10 mm (e.g. Pt/Al_2O_3).
- We shall consider three main types of solid catalyst.
  - Metals
  - Metal oxides and sulfides
  - Zeolites

3.1.1 Metals

- There are many examples.
  - Fe for $N_2 + 3H_2 \rightarrow 2NH_3$
  - Cu/ZnO for $CO + 2H_2 \rightarrow CH_3OH$
  - Ag for $2C_2H_4 + O_2 \rightarrow 2C_2H_4O$
  - Pt for cyclohexane to benzene.
- Metal surfaces are extremely reactive.
  - Many species can exist on them that would be highly unstable in their absence.
  - These species are the intermediates involved in the reaction mechanism.
  - Metals often catalyse more than one reaction – selectivity can be a problem.
- Most commonly the metal is in the form of very small particles dispersed on a porous support material (e.g. forms of Al_2O_3, SiO_2, TiO_2); these are "supported-metal catalysts".
  - The very small metal particles (usually nm size) mean that many metal atoms are on the surface – there is a high metal surface area.
  - The support helps keep the metal nanoparticles separate from each other.
  - The support needs to be thermally stable, robust, and porous so that reagents can reach the metal (and products can transport away).
  - The support is usually, but not always, inert.
- Only occasionally is there no support.
  - Nickel can be prepared in such a way that it is has a high surface area: Raney nickel is used for hydrogenation reactions

- Platinum gauze (a mesh of platinum wire knitted together) is used for $NH_3 \longrightarrow NO_2$ in nitric acid manufacture

- Sometimes the metal is coated on a non-porous monolithic support.
  - Monoliths consist of many parallel channels with walls that are typically extruded ceramics that are then coated with an active ingredient.
  - Monoliths tend to be used for very active catalysts (so surface area is not critical), when low pressure drops are required and heat removal is a problem (e.g. very exothermic reactions)
  - An example is the honeycomb monolith for CO and NO_x conversion in car exhausts.
- The metals used in heterogeneous catalysis often contain commercially sensitive additives.
  - Promoters increase activity – e.g. K is present on Fe for ammonia manufacture.
  - Inhibitors reduce activity but increase selectivity – e.g. Cl is present on Ag for ethylene oxide manufacture.

3.1.2 Preparation of supported-metal catalysts

- There are two main methods for preparing supported-metal catalysts.
- (1) Impregnation
  - Take the porous support and soak it with a solution containing a metal precursor, e.g. Ni^{2+}, $[Pt(NH_3)_4]^{2+}$, $[PtCl_6]^{2-}$
  - There will be ion-exchange between ions in the solution and any ions present on the surface (such as H^+ or OH^- on a support such as Al_2O_3).
  - Dry to give a solid containing the metal precursor.
  - Calcine (heat to high temperature) to cause the chemical change that forms the metal. For instance, calcination will remove NH_3 if $[Pt(NH_3)_4]^{2+}$ is the precursor, or remove HCl gas if $[PtCl_6]^{2-}$ is the precursor. ($MX + O_2 \rightarrow MO_x + X$)
  - Before use, the catalyst will usually need to be activated by heating in H_2 to reduce any passivating layer of oxide on the metal surface.

- (2) Co-precipitation
  - Form a solution containing salts of both the support and the active metal.
  - Precipitation can be induced by a change in temperature, pH or evaporation.
  - When particles of the support begin to form, they act as nucleation sites for metal species.
  - The result is very small particles containing a mixture of phases.
  - Drying, calcination and activation are then performed to give metal particles that are highly dispersed on the support.
  - Example: coprecipitation is used to make Cu nanoparticles intimately mixed with Zn/ZnO particles on a phase of porous Al_2O_3; this is the catalyst for methanol synthesis from syngas.

3.1.3 Metal oxides and metal sulfides

- There are many examples.
  - V_2O_5 for $SO_2 \rightarrow SO_3$ in the manufacture of sulfuric acid
  - CoS/MoS_3 on Al_2O_3 for hydrodesulfurization
- Metal oxides (or sulfides) catalyse reactions by undergoing surface redox reactions.
  - For instance, a simplified mechanism for CO oxidation over copper oxide is

    $2CuO + CO \rightarrow Cu_2O + CO_2$

    $Cu_2O + ½O_2 \rightarrow 2CuO$
- Even if the metal cannot easily change oxidation state (e.g. ZnO), redox reactions can still occur at the surface if the compound is non-stoichiometric (p-type oxides contain some metal vacancies; n-type oxides contain some oxygen vacancies).
- Metal oxides/sulfides are far less reactive than pure metals… but they often catalyse more selective transformations.

3.1.4 Zeolites

- Solid acids and bases can catalyse reactions. The most common type of catalyst in this category are zeolites.
  - Other acidic solids exist, and some solid bases, but they are far less important commercially so not considered further in this course.
- Zeolites are porous crystalline aluminosilicates.
  - A general formula for a zeolite is

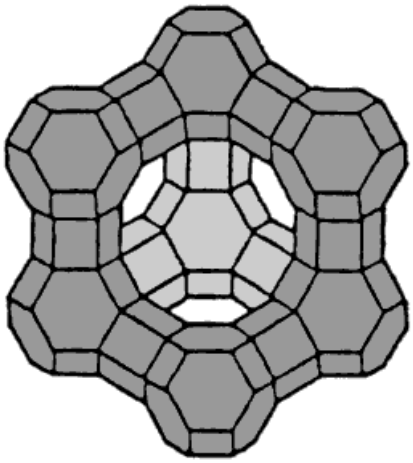
    $(M^{n+})_{y/n}\ [xSiO_2.yAlO_2]$

As acidic solids, they can be poisoned and deactivated by the presence of bases (S, N, H_2O)
Higher Al/Si ratios increase the acidity and the catalytic activity but worsen thermostability and resistance to water (important for catalytic cracking)

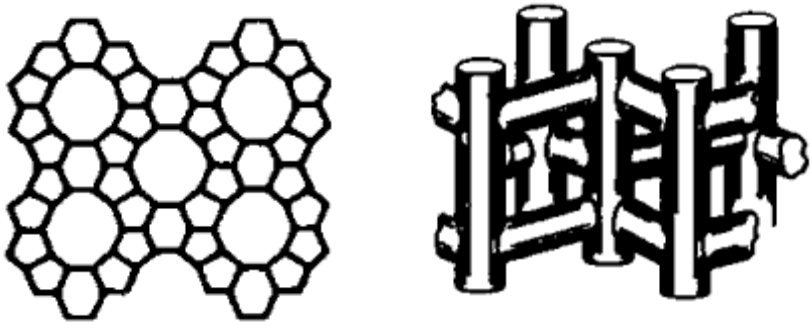
- The framework consists of SiO_4 and AlO_4 tetrahedral units linked to each other.
  - Each aluminium atom contributes a –1 charge to the framework.
  - The framework charge is balanced by cations M^{n+} in non-framework locations.
  - If the cation is H^+, then zeolites have Brønsted acid activity (hydrogen ion donor)
  - If the cation is polyvalent (e.g. La^{3+}), then zeolites have Lewis acid activity (lone pair acceptor).
- Zeolites have pores of molecular dimensions; those used in catalysis have pores just slightly larger than a molecule of benzene.
  - They have high internal surface areas (500 m^2/g in some cases).
  - Virtually all adsorption and catalysis takes place on the internal surface (unless the molecule is too large to enter the zeolite).
- The framework structure may result in
  - channels running through the structure; the channels may be intersecting or non-intersecting
  - cages of larger dimensions which are accessed through smaller pores.
- Diagrams of the framework structure usually just show the Si or Al atom locations. They omit the O atoms and cations for the sake of simplicity.

Framework structure of zeolite Y
(also referred to as faujasite)

This has a cage structure (diameter 2.5 nm)
accessed through smaller pores (diameter 0.75 nm)



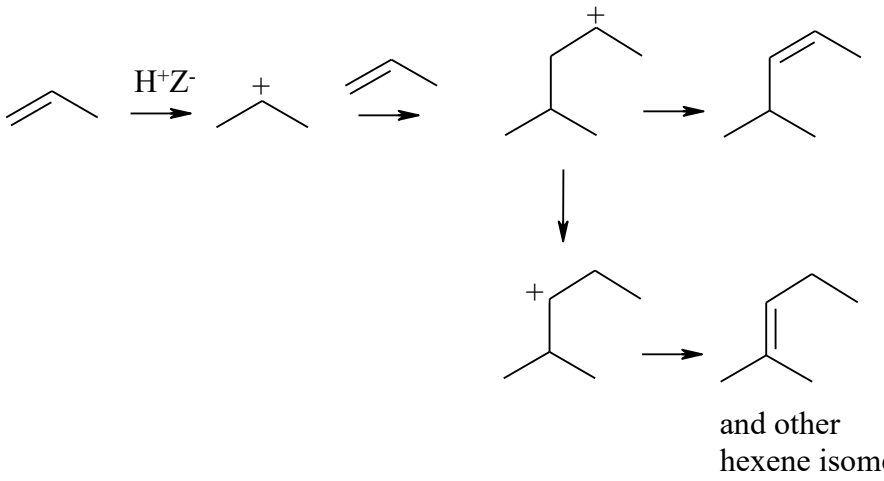
Framework structure of zeolite ZSM-5
This has a network of intersecting channels (approximate diameter 0.56 nm)



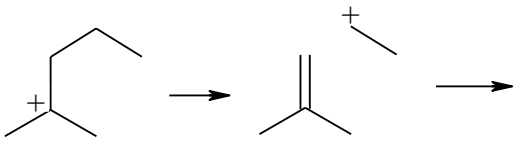
3.1.5 Mechanism for reactions within zeolites

- The intermediates of reactions in zeolites can be cations.
- The following reactions are some of those that will take place when propene reacts in an acidic zeolite.

Dimerisation:



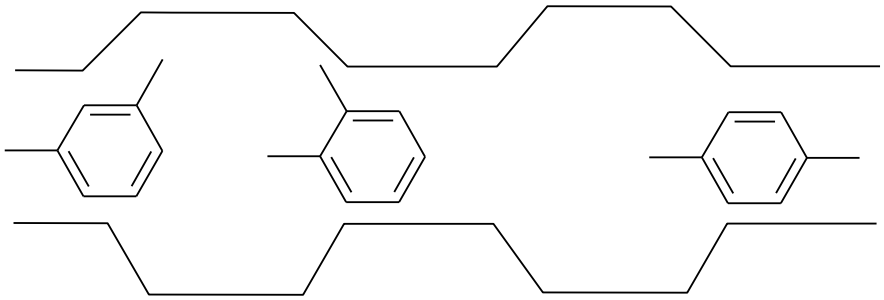
Cracking:



C_4 and C_2 alkanes and alkenes
(similarly get C_1 and C_5 species)

- As well as dimerisation reactions (molecules getting larger), and cracking reactions (molecules getting smaller), there will also be:
  - isomerisation reaction (e.g. *m*-xylene becoming *p*-xylene)
  - cyclisation reactions (e.g. *n*-hexene becoming cyclohexane)
  - dehydrogenation reactions (e.g. cyclohexane becoming benzene).
  - coke formation reactions (e.g. formation of large polyaromatic molecules that become trapped inside the pores of the catalyst)
- Hence it can be seen that a wide range of products will form.

- One reason for the use of zeolites for hydrocarbon transformations is that the pore structure creates selectivity:
  - Sometimes only certain reagents can diffuse into the zeolite to reach the active site (reactant selectivity)
  - Sometimes only certain products can form, as there is insufficient space for intermediate to form that give rise to other products (transition state selectivity)
  - Sometimes only certain species formed within the zeolite can diffuse out to become products (product selectivity).
- Example: Zeolite ZSM-5 is used commercially to produce *p*-xylene from a mixture of xylene isomers. The xylene isomers can interconvert in the larger pore spaces (the intersections between channels), but only the *p*-xylene isomer can easily diffuse out of the zeolite.



- Zeolites can sometimes be tailored to meet the requirements for a particular reaction.
  - The zeolite structure chosen determines the pore size, the pore type (e.g. channels or cages) and the pore connectivity (e.g. intersecting channels or one-dimensional channels).
  - The number and strength of the acid sites for a particular zeolite structure type can often be adjusted by manufacturing conditions, or post-synthesis modification.

3.1.6 Summary of reactions for hydrocarbons on catalysts

Type of catalyst	Example reaction	Example catalyst
Metals	Hydrogenation Dehydrogenation Complete oxidation Hydrogenolysis	Transition metals on right-hand side of periodic table such as Fe, Ni, Cu, Pd, Pt, Ag
Metal oxides;	Selective oxidation	NiO , ZnO,

Metal sulfides	Selective reduction Desulfurization	MnO_2 , MoO_3 , V_2O_5 WS_2, MoS_2
Solid acids	Cracking Isomerization Alkylation Polymerization	Zeolite Y Zeolite ZSM-5

3.2 Catalyst Loss and Deactivation

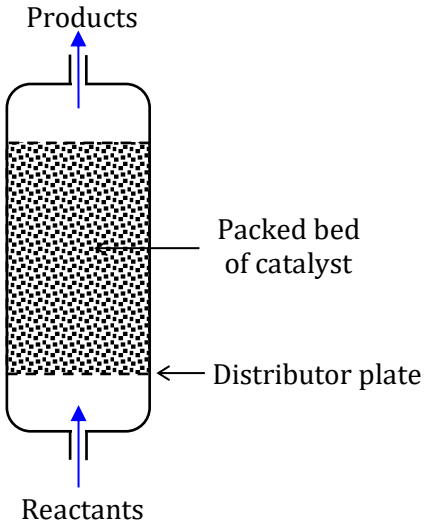
- The following are the three main types of catalyst deactivation.
- Sintering (or aging) – formation of large metal particles from small metal particles.
  - Metal atoms on the surface metal are in a higher energy state than bulk atoms because they do not have their preferred coordination number of atoms around them – there is a thermodynamic driving force for sintering (analogous to bubbles or droplets growing).
  - Very small metal particles have significantly lower melting points than large metal particles. At high temperatures (but below the normal melting point), the mobility of surface atoms means that small metal particles can agglomerate to form larger metal particles.
  - Sintering causes a reduction in metal surface area, and thus in the number of active sites.
- Poisoning – impurities in the feed may permanently occupy reaction sites, so causing irreversible loss of activity.
  - For instance, any sulfur atoms present bind strongly to most metals. (e.g. blackening of sliver)
  - For instance, trace vanadium species in crude oil may build up in zeolites.
- Coking (or fouling) – blocking of the pores or active sites by a reactant or by-product.
  - "Coke" is carbonaceous deposits formed during hydrocarbon transformations.
  - The catalyst may be regenerated by burning off the coke by heating in air – however, this is a very exothermic reaction and the high temperature may adversely affect the catalyst e.g. by causing sintering.
- In addition to these methods of deactivation, catalyst particles can suffer mechanical damage:
  - Impact in fluidised beds.
  - Internal stresses (e.g. arising from temperature profile within pellet) can cause fragmentation. See also capillary condensation (Section 5).

- Dust formation (to give "fines") results in loss of catalyst (and contamination of product). Increase of pressure drops in fixed beds can also result.

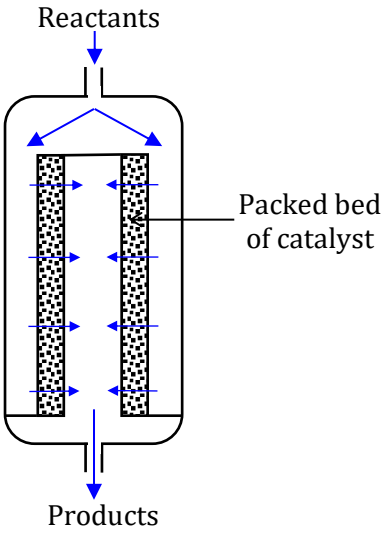
3.3 Types of reactor for heterogeneous catalysis

3.3.1 Fixed bed reactor (also called packed bed reactor)

- This is the most common type of reactor for both vapour-phase and liquid-phase reactions over a solid catalyst.
- The fluid simply passes through a packed bed of catalyst,

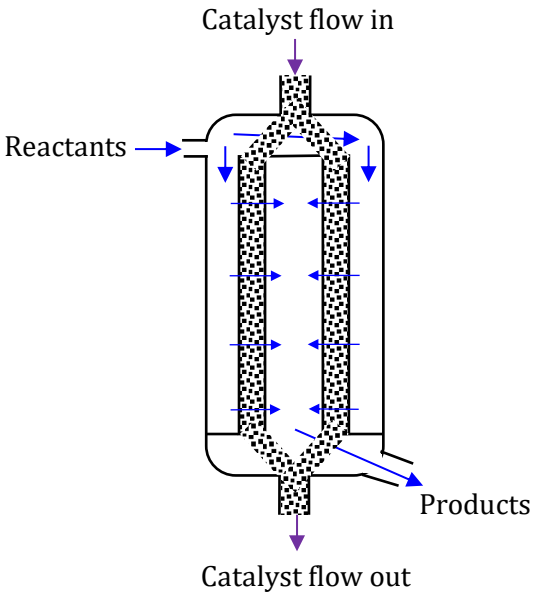


- A fixed bed reactor is a very simple design if they are adiabatic.
  - There will be a temperature gradient along the reactor.
  - If the temperature change would be large, it is common to have more than one adiabatic fixed bed reactor in series with interstage heating/cooling.
- If cooling/heating within the reactor is required (e.g. for highly exothermic reactions):
  - Either there are many tubes containing heat-exchanging fluid that are passed through the bed (e.g. ammonia and methanol synthesis).
  - Or flow of process fluid is through catalyst in tubes that are surrounded by heat-exchanging fluid (a multitubular fixed bed reactor e.g. Fischer-Tropsch synthesis)
- Another configuration is a radial flow reactor which may be used if a small bed length is desired (e.g. low residence time desired and/or if there is little pressure drop available).



3.3.2 Moving bed reactor

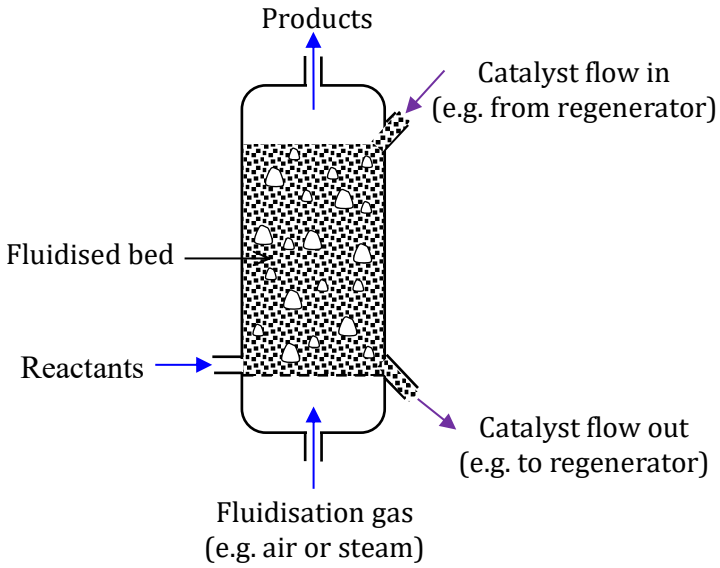
- In a moving bed reactor, some used catalyst is continually removed from the process while fresh catalyst is added.
- This type of reactor may be needed if catalyst deactivation is significant.



3.3.3 Fluidised bed reactor

- This is far less common than a fixed bed of catalyst.
  - Advantage: good mixing; good temperature control.
  - Disadvantage: stringent operating conditions for flowrates and particle characteristics (size, shape, robustness).

- If some catalyst is added and withdrawn continuously from the reactor, then this is a type of moving bed reactor.



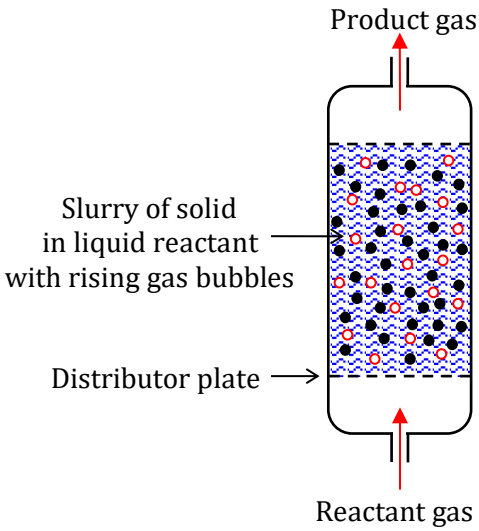
- Example: fluidised catalytic cracking (FCC) units on oil refineries.
  - These use a zeolite-based catalyst.
  - The cracking reactions cause coke to build up within the catalyst particles in about one second under operating conditions.
  - Catalyst is continuously sent to a regenerator unit which is itself a fluidised bed. The regenerator has an air feed and operates at high temperature (800°C) to burn off the coke.

3.3.4 Multiphase reactor

- Some reactions involve gas and liquid as well as a solid catalyst.
  - Sometimes both gas and liquid are reagents, e.g. H_2 and a hydrocarbon liquid with a hydrodesulfurisation catalyst (e.g. diesel desulfurisation)
  - Sometimes the liquid is simply a heat sink to reduce the temperature increase for a highly exothermic reaction.

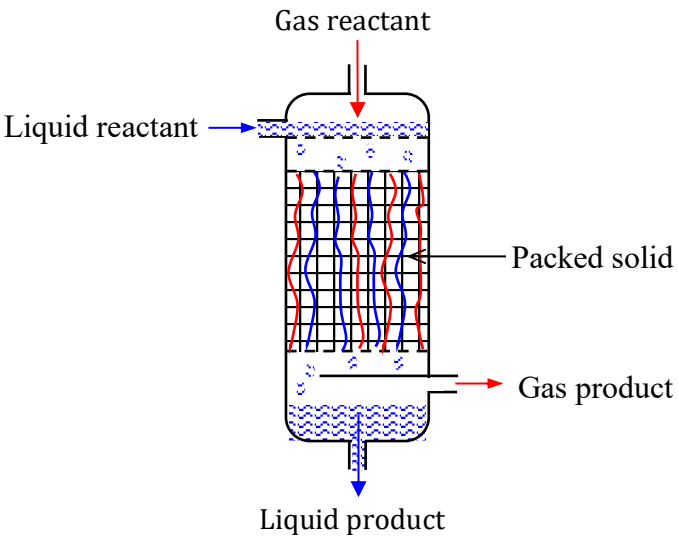
(a) Slurry reactor

- Here gas is bubbled through a suspension of catalyst in the liquid.
- Disadvantage: liquid and solid may need later separation.
- Need CET IIA two-phase flow to describe the bubble size as the gas flows through the liquid.



(b) Trickle bed reactor

- Here gas and liquid flow (usually) cocurrently downward through a fixed bed of catalyst.
- There are different flow regimes depending on the gas and liquid flowrate, the particle size and the particle properties.



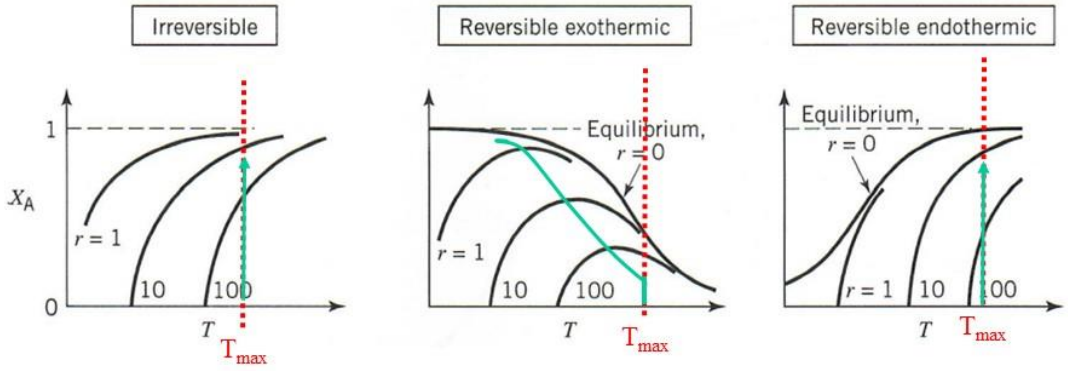
3.4 Modelling heterogeneous reactions

- A fixed bed reactor might approximate a PFR (though not very well).
- A fluidised bed reactor might approximate a CSTR (though not very well).
- However, we need to understand both mass transport and chemical kinetics to predict the form of the overall rate law (and thus the conversion).
- Often the main difficulties of design of catalytic reactors (solid catalyst / fluid)
  1) Non-isothermal behaviours in packed beds
  2) Non-ideal flow in fluidized beds

3.4.1 Staged adiabatic packed bed reactors

Assume each reactor is an ideal PFR

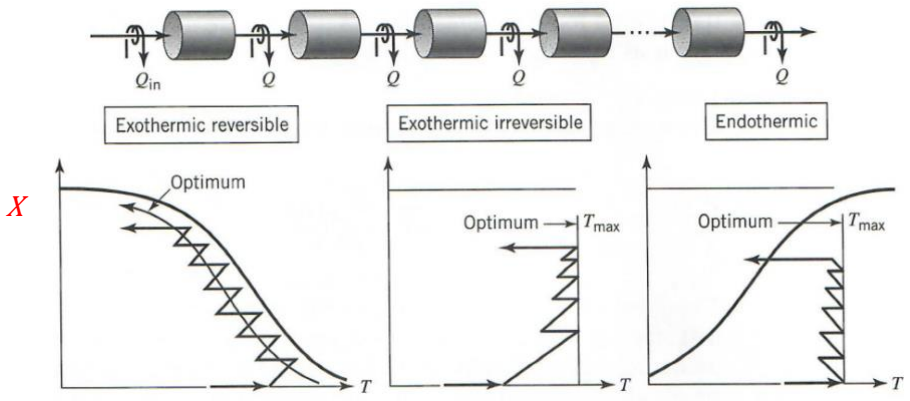
- We can build plots, reporting the conversion as a function of the temperature, at different reaction rates.
- Given the maximum operating temperature (technological limitations, materials' properties, etc…) the optimal operating temperature profile to minimize the reactor size is the locus of maximum rates (green).



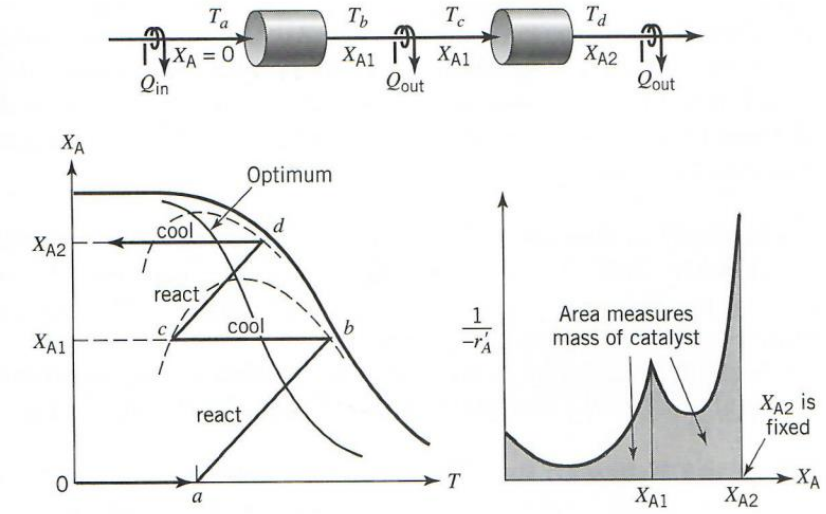
- For adiabatic operations we know that: $X_A = \frac{c_p(T-T_0)}{-\Delta H_r}$

ΔH_r has a weak dependence on T

- Adiabatic stages can be used to approximate the ideal temperature profile



- The volume of each stage can be easily calculated plotting $-\frac{1}{r_A}$ vs. X_A.



3.4.2 Packed bed and trickle bed reactors

- We can approximate packed beds and trickle beds as N equivalent CSTRs.
- The geometry of our system gives an indication of N.
  - N may be the average number of spaces between particles along the length of the bed.
  - For single-phase flow at high Reynolds numbers in packed beds $N \approx L/d_{\text{particles}}$ if there is good mixing in the space between particles.
    [L = length of bed; $d_{\text{particles}}$ = characteristic size of particles]
  - For slow single-phase flow in packed beds it is sometimes found that $N \approx L/4d_{\text{particles}}$ is reasonable.
- Alternatively we can measure the RTD and choose N so that the variance of the RTD for the model ($\sigma^2 = \tau^2/N$ from Section 1.4.1) agrees with the experimental measurement.
- The single parameter N quantifies where on the scale from ideal CSTR to ideal PFR the overall reaction vessel lies.

- If there is chemical reaction, we can now do concentration calculations within each of the CSTRs to determine the overall conversion in the system.

Example – BP diesel desulfurisation process

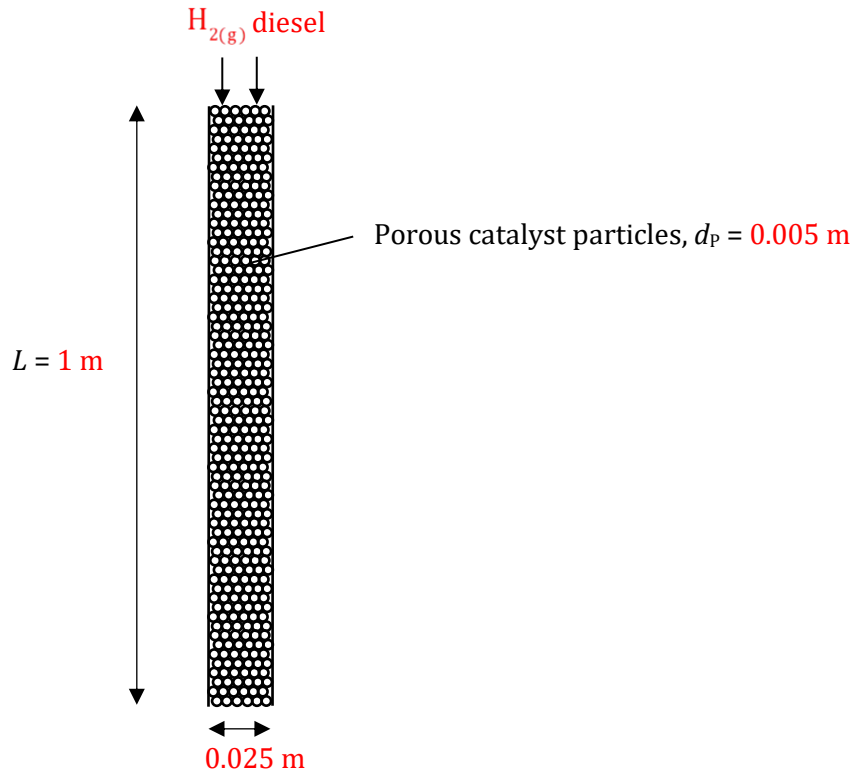
(a real example, but some numbers invented, since I don't know the catalyst particle size)

The reaction is:

$$H_{2\,(g)} \xrightarrow{\text{dissolves in diesel}} H_{2\,(l)} + RSH_{(l)} \xrightarrow{\text{Pt catalyst}} H_2S_{(l)} + RH_{(l)}$$

The reactor described operated in a pilot plant, where studies (primarily catalyst development) were made prior to the implementation of changes on the full-scale plant.

The reaction occurs in a trickle bed reactor with co-current flow:



The process operates at 50 bar and 350°C. At this pressure H_2 is highly soluble in diesel. Also sulfur impurities in diesel occur at relatively low concentration. Thus the reaction may be considered pseudo-first order in concentration of RSH:

$-r = k'[\text{RSH}]$ per unit volume, i.e. for a specific catalyst morphology and packing

with $k' = 0.0552\ \text{s}^{-1}$

[Note: this is how heterogeneous reactors are analysed. An "average rate constant" is defined, and the reactor is then treated as if it were homogenous. All concentration and temperature variations within the reactor are concealed in this "average rate constant". Thus k' is no longer really a constant, but can vary with flow rate, particle size, porosity, catalyst age, etc.]

The plant ran happily at 90% removal of sulfur from diesel, and had always been successfully modelled as a PFR.

The regulations changed to require 99.9% sulfur removal from diesel.

Consider the PFR model:

Design equation: $F_{A0}dX = -rdV$

$= k'C_A dV$ A = RSH

$= k'C_{A0}(1-X)dV$

$$\frac{F_{A0}}{C_{A0}}\int_0^X \frac{dX}{1-X} = k'V$$

$Q\ln(1-X) = -k'V$ $\frac{V}{Q} = \tau$ assuming constant density

$\ln(1-X) = -k'\tau$ assuming plug flow operation

Originally, $X = 0.9 \Longrightarrow k'\tau = 2.3 \Longrightarrow \tau = 42$ s

Check the Reynolds number: $u = \frac{L}{\tau} = \frac{1}{42} = 0.024$ m s^{-1}

The kinematic viscosity of diesel at 350°C $\approx 10^{-5}$ m^2 s^{-1}

$$\Rightarrow Re = \frac{ud_P}{\upsilon} \approx 12$$

Re is low – so expect mixing in inter-particle spaces to be poor, and so the reactor is not likely to behave much like an ideal PFR. This will be enhanced by the reactor being a trickle bed – the gas phase tends to disperse the liquid phase.

Now we require X = 0.999

$\ln(0.001) = -k'\tau$ assuming PFR

$k'\tau = 6.9 \Rightarrow \tau = 125$ s

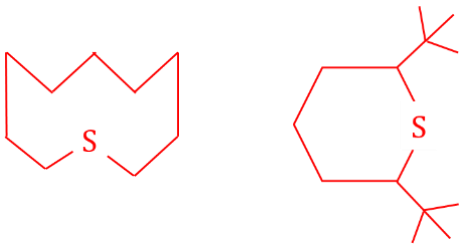
So we should simply be able to treble the residence time to achieve the required conversion.

On doing this, by reducing the liquid flow rate to one third its original value, keeping everything else the same, BP found that the sulfur content was still **5 times the permitted level!**

WHY?

The initial thought was that it was due to steric hindrance in a portion of the sulfur containing molecules, the idea being that these would have a lower rate constant for reaction and so be particularly slow to remove.

e.g.



It is difficult for the S atoms to access the catalyst surface in such molecules, resulting in slow reaction.

After extensive (and expensive) analysis of the nature of the sulfur containing molecules and measurement of their rate constants, it was shown that this could not be the explanation.

In reality the problem lies with the assumption of plug flow. The reactor will not behave as an ideal PFR because of axial dispersion caused by mixing in the inter-particle spaces and dispersion of liquid by gas. This can be modelled as a cascade of CSTRs.

How many CSTRs?

The best thing to do would be to measure variance of RTD (by injection of a tracer pulse) and match to model variance. This was not done.

Recall simple guesstimate for packed beds:

$$N = \frac{L}{d_P}$$

and at low Re:

$$N = \frac{L}{4d_P}$$

In trickle beds, the presence of the gas phase promotes dispersion, so we need even fewer CSTRs in the model.

The case of trickle beds is more complex than simple packed beds, but at low Re, co-current trickle beds can be modelled using:

$$N = \frac{L}{20 d_P}$$

[see J.M. Hochman and E. Effron (1969) I&EC Fund 8, 63; W.P.M. van Swaaij et al. (1969) Chem. Eng. Sci. 24, 1083; D.E. Mears (1971) I&EC PDD 10, 541 for details]

Thus we have:

$$N = \frac{L}{20 d_P} = \frac{1}{20 \times 0.005} = 10$$

So we should be able to model this trickle bed reactor as a chain of 10 CSTRs.

Remember that for a **first order** reaction degree of **mixedness** at the molecular level is **irrelevant** (analogy with radioactive isotope decay). Thus regardless of the real state of molecular mixedness, we can choose to model the reactor as either macro-mixed or micro-mixed and we will get the same answer.

Let's do the calculation for the micro-mixed limit:

Mass balance over a single CSTR:

$$Q(C_{A0} - C_A) = -r_A \frac{V}{N} = k' C_A \frac{V}{N}$$ ($\frac{V}{N}$ is the volume of a single CSTR)

$$N\left(1 - \frac{C_A}{C_{A0}}\right) = k'\tau \frac{C_A}{C_{A0}}$$ (τ is the residence time of the whole reactor, i.e. N CSTRs)

$$\frac{C_A}{C_{A0}} = 1 - X_{CSTR} = \frac{N}{N + k'\tau}$$ (X_{CSTR} is the conversion over a single CSTR)

Note that this is independent of C_{A0}, so the conversion is independent of the inlet concentration for all CSTRs in the cascade (this is because it is a first order reaction).

Hence for N CSTRs:

$$1 - X = (1 - X_{CSTR})^N = \left(\frac{N}{N + k'\tau}\right)^N$$ (X is the conversion over the whole reactor. i.e. N CSTRs)

[Aside: Repeat the calculation for the macro-mixed limit to check that we get the same answer:

Use the mini-batch reactor analogy:

$$1 - X = \frac{C_{\mathrm{A}}}{C_{\mathrm{A0}}} = \frac{1}{C_{\mathrm{A0}}} \int_0^{\infty} C_{\mathrm{A}}^{\mathrm{batch}} E(t)\, dt$$

For N CSTRs in series (Section 1.4.1),

$$E(t) = \frac{t^{N-1}}{(N-1)!\,\tau_i^N} \exp\left(\frac{-t}{\tau_i}\right)$$

For 1st order reaction in a batch reactor:

$$\frac{dC_{\mathrm{A}}^{\mathrm{batch}}}{dt} = -k' C_{\mathrm{A}} \Rightarrow \ln \frac{C_{\mathrm{A}}^{\mathrm{batch}}}{C_{\mathrm{A0}}} = -k't$$

$$C_{\mathrm{A}}^{\mathrm{batch}} = C_{\mathrm{A0}} \exp(-k't)$$

So for a 1st order reaction, in any continuous reactor:

$$1 - X = \frac{1}{C_{\mathrm{A0}}} \int_0^{\infty} C_{\mathrm{A0}} \exp(-k't)\, E(t) dt \qquad \text{(N.b. independent of } C_{\mathrm{A0}}\text{)}$$

For a cascade of 10 CSTRs:

$$1 - X = \frac{1}{9!\,\tau_i^{10}} \int_0^{\infty} \exp(-k't)\, t^9 \exp\left(-\frac{t}{\tau_i}\right) dt$$

$$= \frac{1}{9!\,\tau_i^{10}} \int_0^{\infty} t^9 \exp\left(-\left[k' + \tfrac{1}{\tau_i}\right] t\right) dt$$

integrating by parts

$$= \frac{1}{9!\,\tau_i^{10}} \left\{ \left[-\frac{1}{\left(k' + \frac{1}{\tau_i}\right)} t^9 \exp\left(-\left[k' + \tfrac{1}{\tau_i}\right] t\right) \right]_0^{\infty} + \frac{9}{\left(k' + \frac{1}{\tau_i}\right)} \int_0^{\infty} t^8 \exp\left(-\left[k' + \tfrac{1}{\tau_i}\right] t\right) dt \right\}$$

… and another 8 times …

$$= \frac{1}{\left(k' + \frac{1}{\tau_i}\right)^{10} \tau_i^{10}} \left[-\exp\left(-\left[k' + \tfrac{1}{\tau_i}\right] t\right) \right]_0^{\infty}$$

$$= \left(\frac{10}{10 + k'\tau}\right)^{10}$$

i.e. the same answer as was obtained for the micromixed limit]

Assuming the reactor to be behaving as a PFR, BP increased $k'\tau$ to 6.9 (from 2.3), aiming to achieve 99.9% conversion.

For $N = 10$,

$$1 - X = \left(\frac{10}{16.9}\right)^{10} = 0.0053$$

This is 5 times the allowable RSH levels – as observed!

In order to achieve 99.9% conversion, we require:

$$\left(\frac{10}{10 + k'\tau}\right)^{10} = 0.001 \Rightarrow k'\tau = 10$$

i.e. we need to increase the residence time by nearly 5 times (instead of 3).

This could be done by reducing the liquid flow rate to one fifth its original value. There is a risk that in doing this, the mixing in the inter-particle spaces would get even poorer, so further reducing the number of CSTRs required in the model (the papers cited above could be consulted to determine this).

A better (but more expensive) method, would be to increase the length of the tube 5 times. If we did this, the number of CSTRs required in the model would go up (also by 5 times) and so the dispersion would go down and the conversion up. Thus we would not require a tube 5 times as long to get the desired improvement (somewhere between 3 and 5 times as long would be necessary). You could use your reactor model to estimate how much longer the tube should be – perhaps you would like to try this?

Conclusion

It can be dangerous to assume that reactors are behaving ideally, particularly when **high conversions** are required. This is because a small amount of material passing through in significantly shorter than expected residence time can make a big difference to the amount of unreacted reactant remaining.

Thus **more sophisticated modelling** of RTD (and mixedness if the reaction is not first order) is required. However **fairly simple models** usually suffice, e.g. cascade of CSTRs.

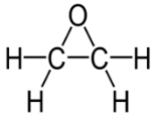
3.5 Some examples of industrial interest

- **3.5.1 Ethylene oxide: a supported metal catalyst**

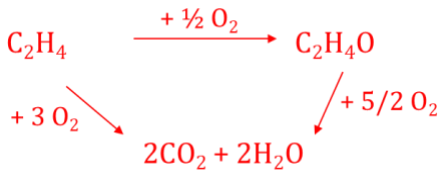
World production ~20 Mt/year.

65% is used to make ethylene glycol, which is used in antifreeze (49%) and polyesters (45%).

13% is converted to surfactants, 7% to glycol ethers and 6% to ethanolamine.



- It is produced by catalytic partial oxidation of ethylene. Complete oxidation of ethylene and ethylene oxide can reduce the selectivity



- The catalyst is Ag (5-30%) supported on Al_2O_3.
- It is prepared by impregnation of Ag amines complexes.
- A flow of warm N_2 is used to dry and decompose the complexes.
- Ag^+ is then reduced to metallic Ag by using water steam at 200 °C. This gives good spatial distribution of the active species.

Operational parameters:

- **Ag crystals' dimensions**: larger crystals have a higher purity and less defects in crystalline structure and give a higher selectivity. However, too large crystals decrease the available surface area
- **Support**: Silica gives better selectivity than alumina. The alumina surface has a weak acidic behaviour that can catalyse isomerization of ethylene oxide to acetaldehyde, which is easily converted to CO_2 and water. However, at the operating temperature, silica cannot be used, because it has worse thermomechanical properties.
- Small quantities of promoters (Cs, Li, K, Ba, Na, alkaline metals) are deliberately added to the support to reduce the acidity, increasing the selectivity. These promoters also reduce sintering.
- Sulfur-Ag bonds are very strong, causing the catalyst deactivation.

3.5.2 Methanol synthesis

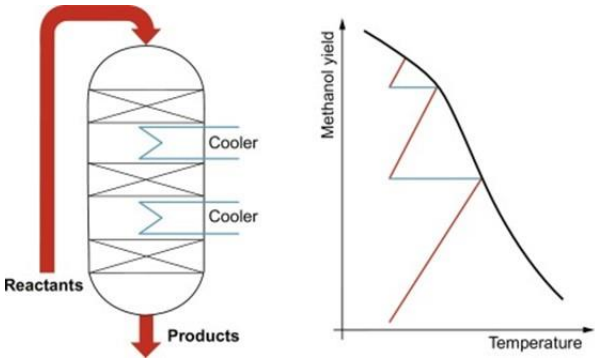
Methanol used in the synthesis of formaldehyde, acetic acid, chloromethane, amines, etc.

$$CO + 2H_2 \rightleftharpoons CH_3OH$$

- The catalyst is $Cu/ZnO/Al_2O_3$
- Coprecipitation is used to make Cu nanoparticles intimately mixed with ZnO particles on a phase of porous Al_2O_3.
- ZnO is a promoter via Strong Metal Support Interaction (SMSI); Al_2O_3 is the support; however the role of each component and the synergistic interaction of Cu and Zn is an active area of research
- Catalyst deactivation can be caused by S and Cl.

One reactor configuration used is:

<u>Staged adiabatic packed bed reactors (axial or radial flux)</u>



3.5.2 Catalytic oxidation of ammonia: a peculiar catalyst design

Ostwald process for nitric acid production starting from ammonia

$$4NH_3 + 5O_2 \rightarrow 4NO + 6H_2O$$

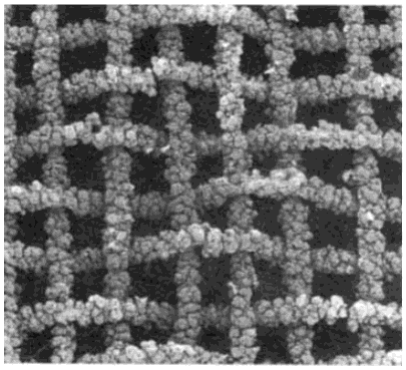
$$4NH_3 + 3O_2 \rightarrow 2N_2 + 6H_2O$$ (competing reaction, thermodynamically favoured at all T)

Pt is active at high T (750-850 °C) with a high selectivity to NO (94-98%)

At this temperature, both reactions are very fast and the competing reaction can occur in the homogeneous phase.

For this reason the catalyst contact time must be extremely low (ms)

- Porous particles cannot be used!
- Platinum wires of about 70 nm (diameter) are knitted together to form a gauze





- Rhodium is used (6-15%) as a promoter, increasing selectivity to NO (up to 99%)
- Rhodium also increases the life of the platinum gauze (Pt is very expensive!)
- Catalyst deactivation/aging is mainly due to:
  - Traces of C_2H_2, H_2S, As, Cl, Si, Fe transported by the gaseous phase.
  - Poisons and high T can corrode the gauze with partial Pt losses (Rh can slow down this phenomenon)
- Losses of Pt can be quantified in mg of Pt per ton of HNO_3 produced. In some plants, low temperature filters are used to partially recover Pt.
- Initial Pt losses of new gauzes cause an increase in catalytic activity due to an increase in surface roughness. However, after a characteristic time, the activity decreases.

3.6 Mechanism

- For reactions within a particle such as a supported metal catalyst, the reaction steps may include:
  1. External mass transport of reactant to the catalyst particle – Section 2
  2. Internal diffusion of reactant within the catalyst particle to reach the active site – Section 6
  3. Adsorption onto the active site (which may be molecular or dissociative) – Section 4
  4. Surface reaction (which may or may not require surface diffusion to occur) to form a product – Section 5
  5. Desorption of the product from the active site
  6. Internal diffusion of product away from the active site
  7. External mass transport of product away from the catalyst particle

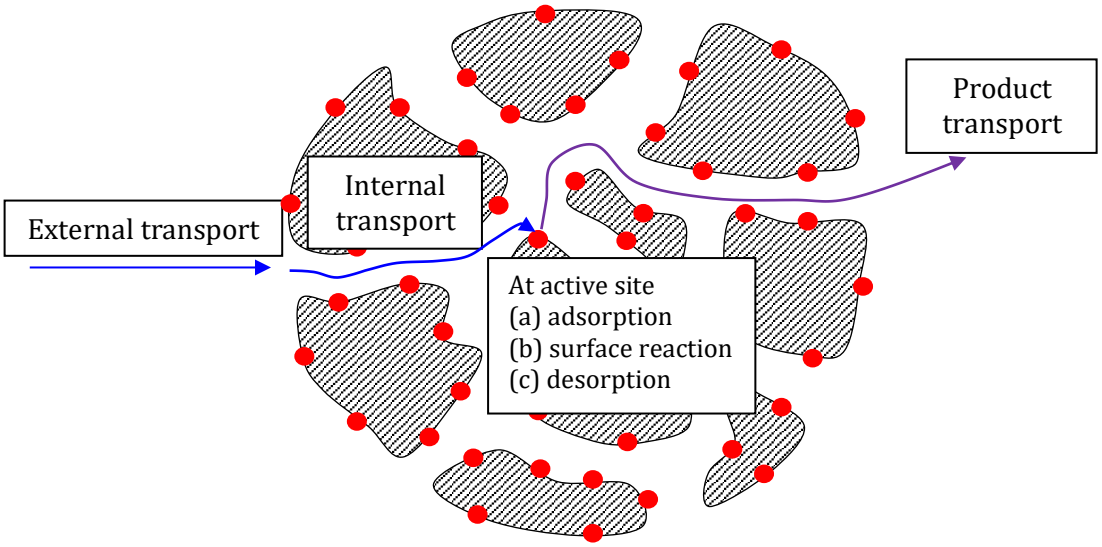


Figure: Schematic of steps involved in reaction within a supported-metal catalyst

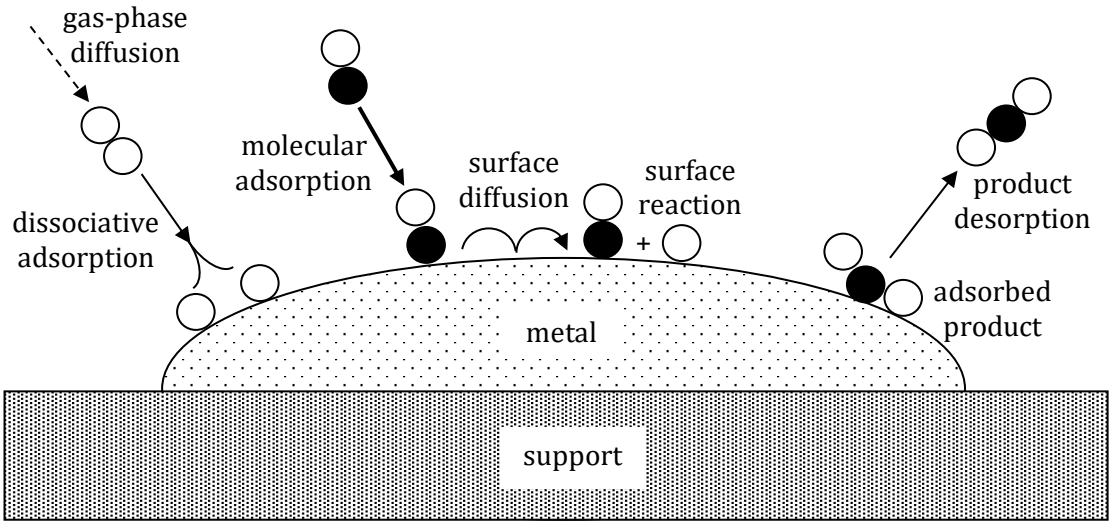


Figure: Schematic of adsorption/reaction/desorption steps

for the reaction ½O_2 + CO → CO_2 on a supported metal catalyst.